Assignment

Digital Image Processing

Course Code: CSE 4141 **Marks: 10**

Answer the following questions :

1. What is the difference between **image restoration** and **image enhancement**? Which one is more objective and why?

2. Why is the **median filter** preferred over linear smoothing filters for removing salt-and-pepper noise? What property of the median makes it robust to outliers?

3. What is the key advantage of an **adaptive filter** over a fixed (non-adaptive) filter in image restoration?

4. How does the adaptive median filter's two-level algorithm (Level A and Level B) decide whether a pixel is a noise spike or genuine image detail?

5. If you were restoring a **medical X-ray image** corrupted by Gaussian noise, which filter(s) from this lecture would you choose and why?

6. How does the **geometric mean filter** preserve more image detail compared to the arithmetic mean filter, even though both compute some form of average?